

Development of a Quantitative Accounting Framework for Black Carbon and Brown Carbon from Emissions Inventory to Impacts

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Framework for Black Carbon and Brown Carbon from Emissions to Impacts
EPA STAR Grant R83503901



Motivation

- Emissions inventories and air quality models of light absorbing carbon require parameterization of the radiative properties of emissions
- Current parameterizations of light absorbing carbon emissions do not address the range of variability within sources or control technologies
- Elemental carbon is not a good surrogate for light absorbing carbon for control strategy development nor assessment of control strategy implementation
 - May be OK if limited to absorption at 880 nm
- The light absorbing capacity of carbonaceous aerosol is not a conservative property from the point of emissions to atmosphere



Project Goals

- Overall Goal
 - Development of a quantitative framework for source-receptor relationships for light absorbing carbon and their associated wavelength dependent light absorptivity
- Key Objectives
 - Deconstruct emissions from sources of light absorbing carbon to elucidate the contribution of different emissions components to wavelength dependent absorption
 - Elucidate how the evolution of emissions in plumes impact wavelength dependent absorption
 - Integrate source apportionment models for aerosol components impacting light adsorption with wavelength dependent light absorption closure calculations



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Project Strategy

- Source Testing
- Mie theory calculations for source emissions and deconstructed emissions
- Atmospheric measurements
- Mie theory calculations for atmospheric aerosols and deconstructed aerosols
- Develop a source apportionment framework that can address the optical evolution of aerosols and precursors



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Source Testing

- Examine key sources of light absorbing carbon:
 - Mobile sources
 - Conventional CI and SI and Emerging Technologies
 - Biomass burning
 - Lab and Field Studies
 - Coal combustion
- Examine for each source
 - Role of process variables on emissions
 - Optical properties of the organic carbon
 - Optical properties of the elemental carbon
 - Impact of dilution
 - Impact of thermal stripping of organics
- Develop source specific light absorption closure models for measurement conditions and high dilution conditions



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Atmospheric Measurements

- Use sites where we have conducted source apportionment studies in the past and where historical record and optical measurements
 - Atlanta, Georgia
 - Near Roadway
 - Rural Alabama
 - SOA
 - India
 - Biomass and Trash Burning
 - Low Temperature Coal Combustion



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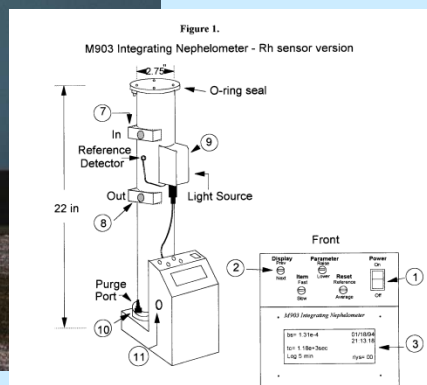
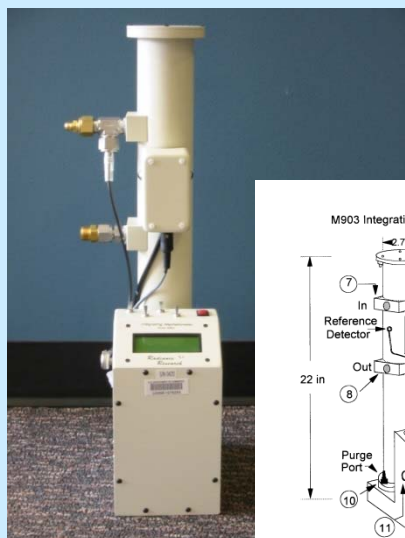


Approach

- Measure the optical properties under controlled conditions
 - Scattering and Absorption (multiple wavelengths)
- Measure physical-chemical properties
 - Size distribution, particle shape, chemical composition
- Segregate components of aerosols
 - Thermal Denuder, WS and Organic solvent atomization
- Correct absorption artifacts and compare optical properties of aerosol components



Methods



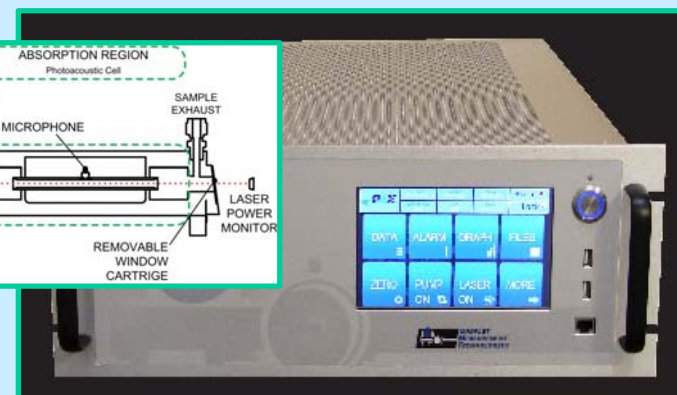
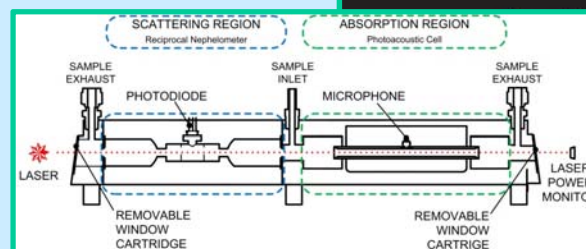
Radiance Research Nephelometer



Magee Scientific AE31 7-channel Aethalometer



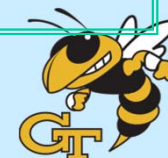
TSI Scanning Mobility Particle Sizer/ Electrostatic classifier



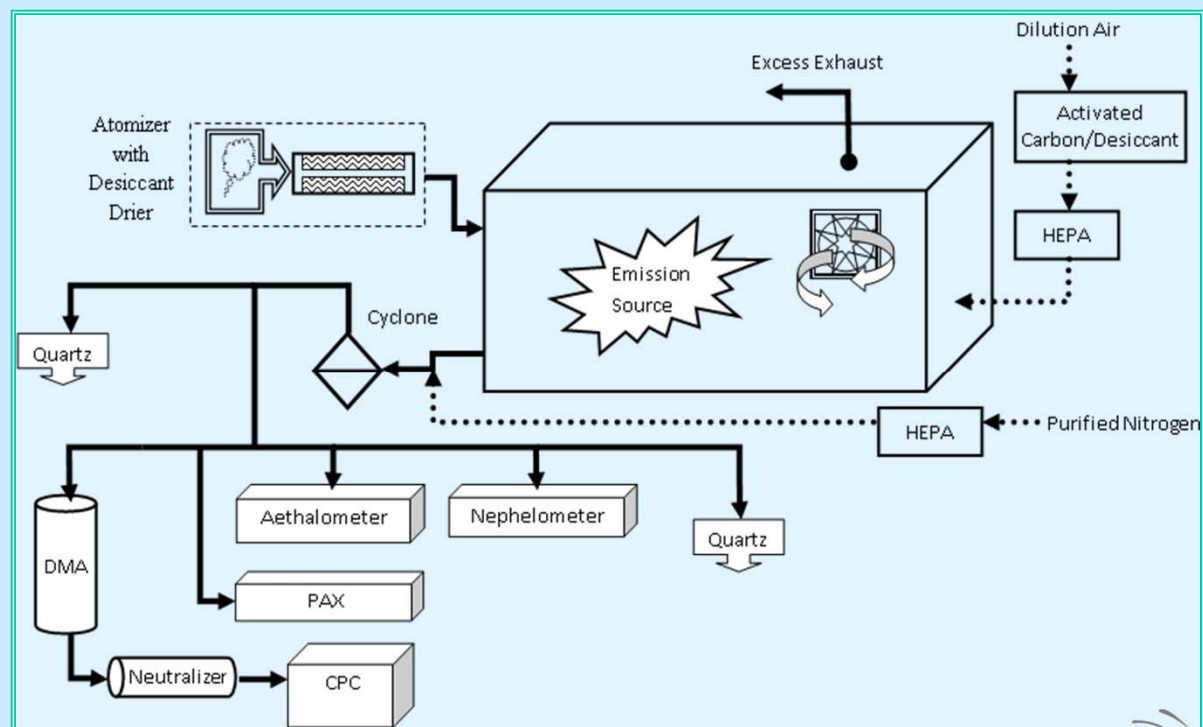
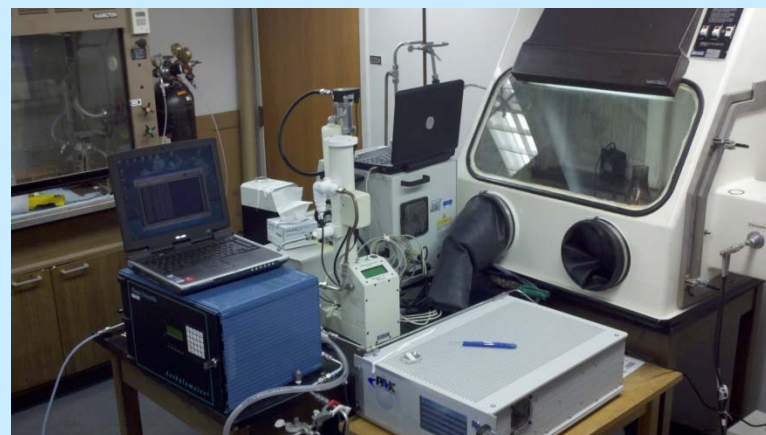
DMT PAX 532: Photoacoustic Extinctionometer



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Methods

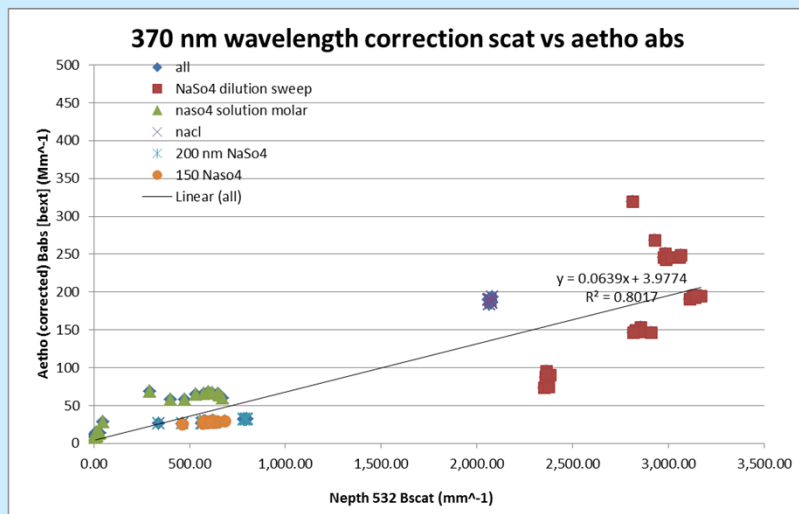


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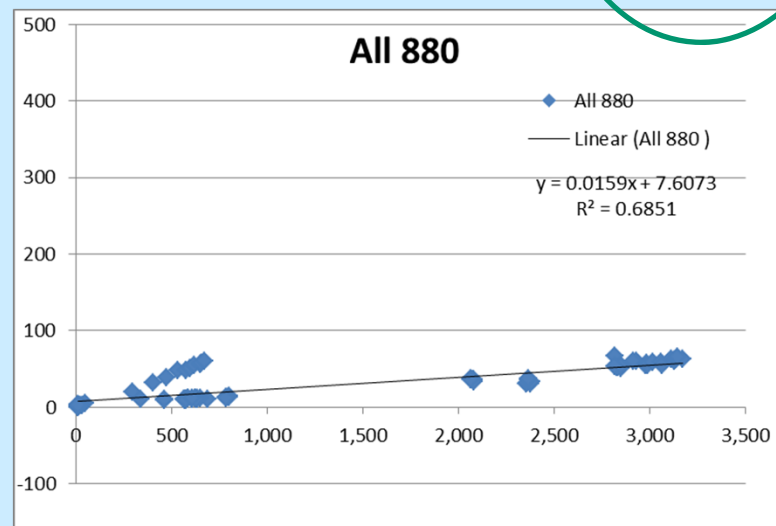
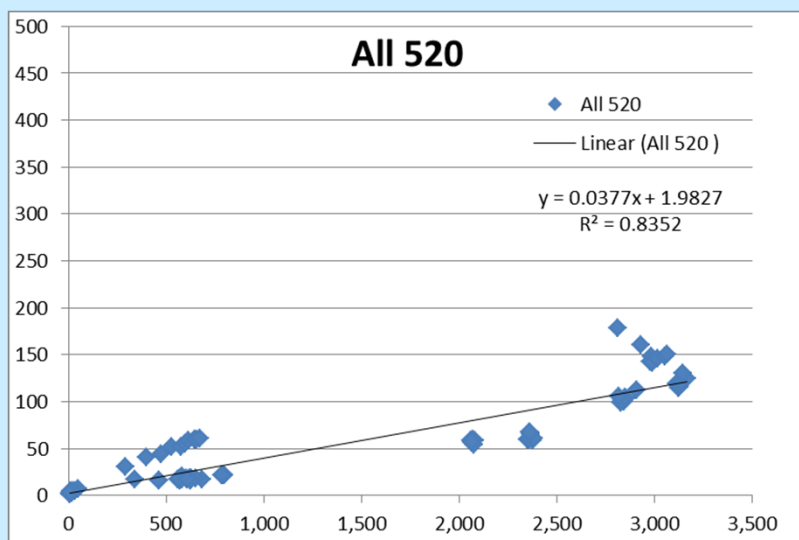


Attenuation by Non-Absorbing Aerosols

Absorption vs Scattering: Scattering Artifact correction



Wavelengths	slope (m)	Intercept (b)	R ²	slope forced through zero (m')
370	0.064	3.977	0.801747	0.066
470	0.049	-0.909	0.816126	0.048
520	0.038	1.983	0.835205	0.039
590	0.030	2.919	0.801001	0.032
660	0.027	3.207	0.84761	0.028
880	0.016	7.607	0.685092	0.019
950	0.013	7.410	0.651662	0.016

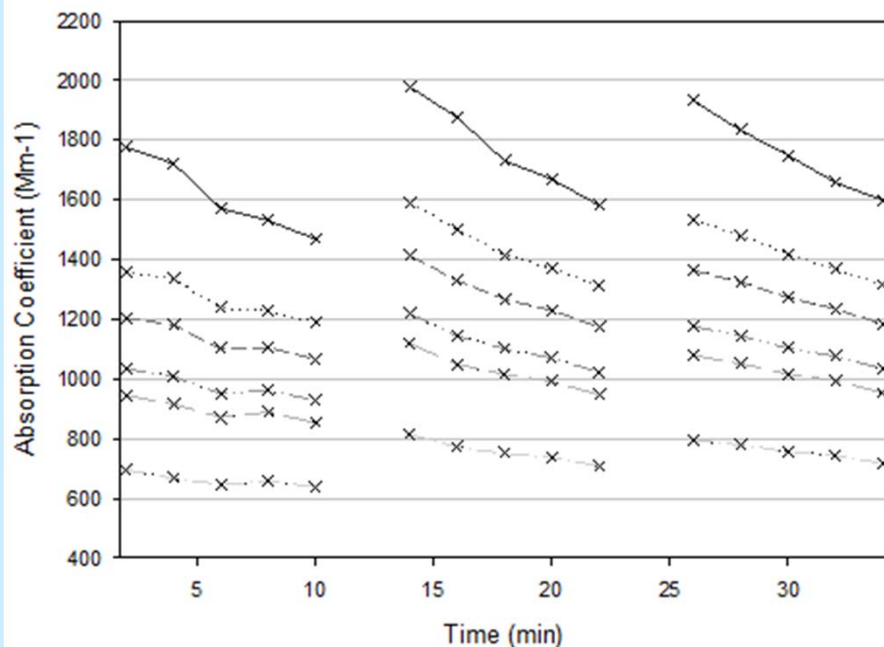


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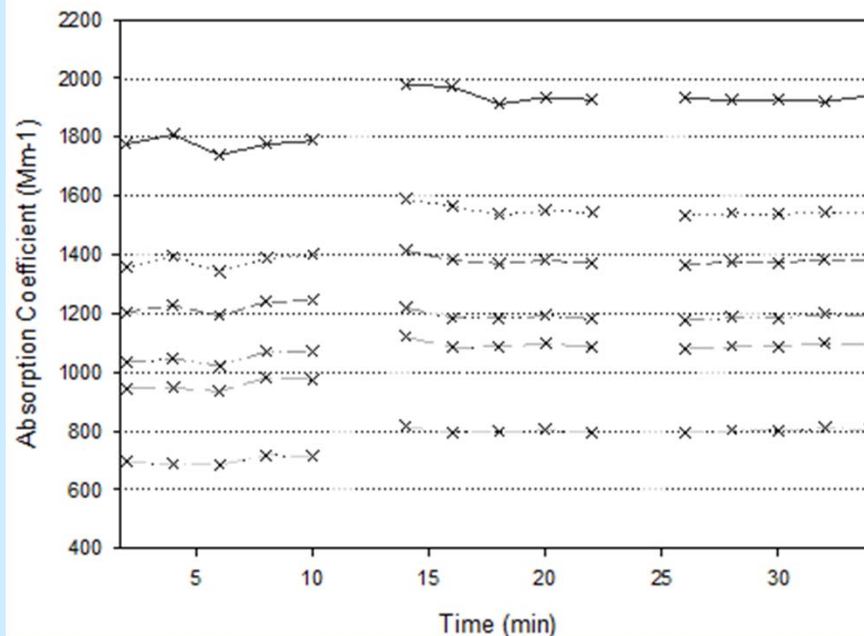


Multiple Wavelength Absorption Correction

Idle Engine Out, TD on, Not Corrected Absorption



Idle Engine Out, TD on, Corrected Absorption



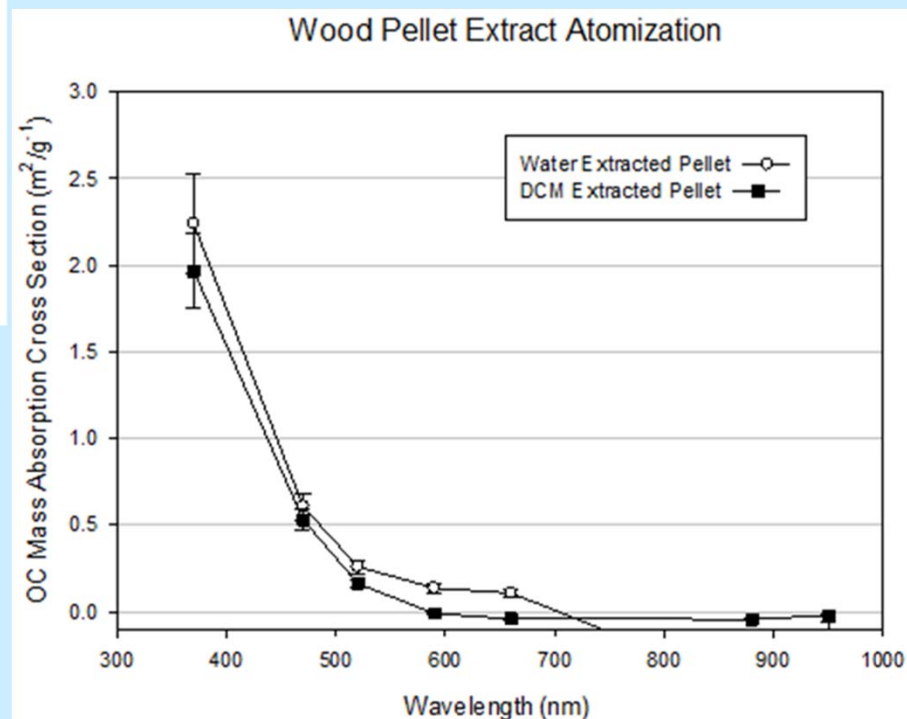
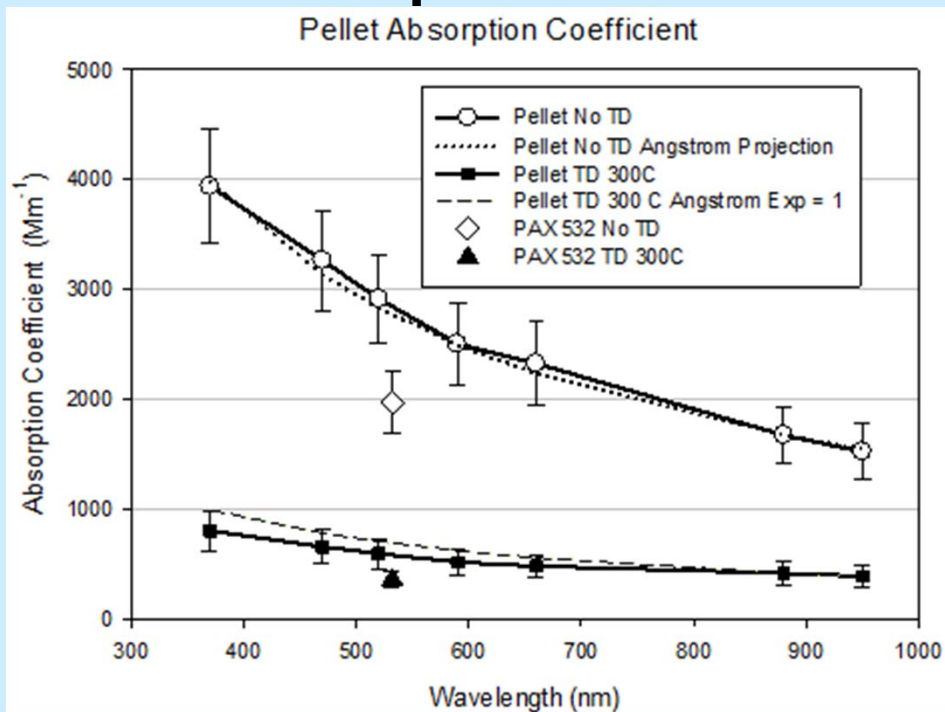
- Test run at steady-state
- Scattering correction is not significant for engine out emissions
- Loading correction is wavelength specific



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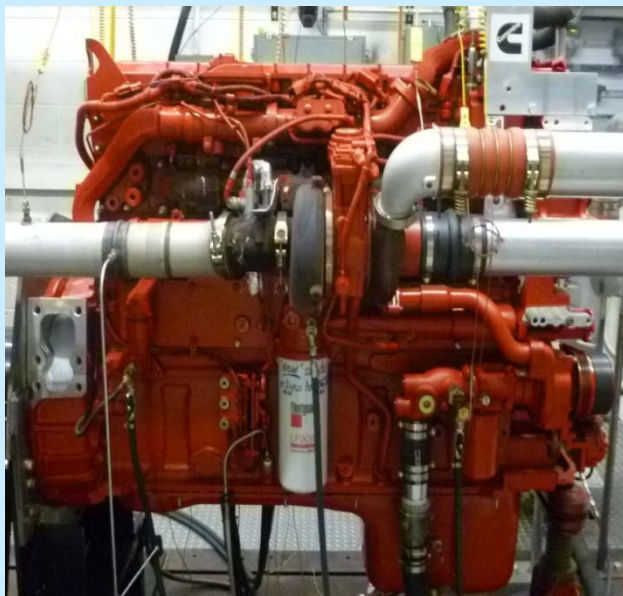


Example of Wood Pellets



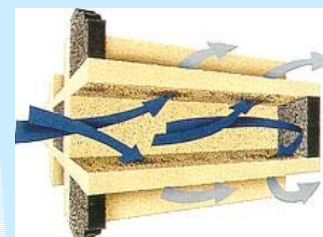
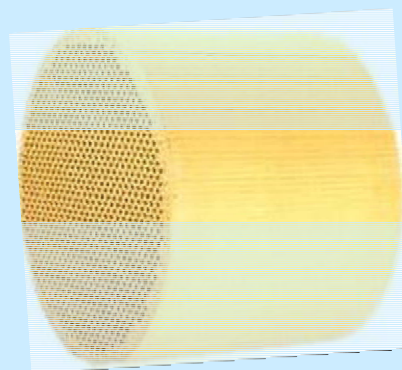
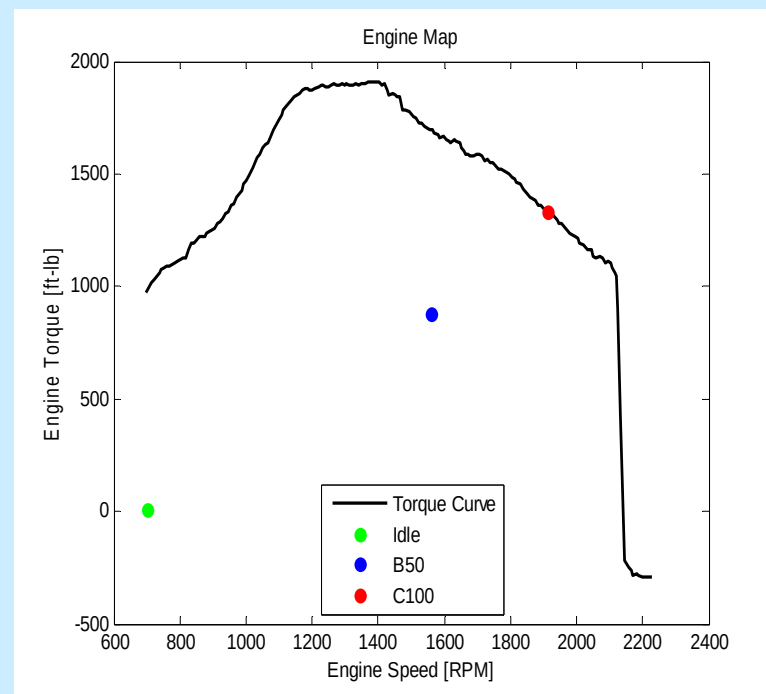
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Experimental Setup

Model	2010, Cummins ISX15 – 500
Emission Certification	EPA 2010, CARB 2010
Type	4-stroke cycle
Cylinder Configuration	In-line 6
Bore and Stroke	137 mm x 169 mm
Compression Ratio	17.2:1
Aspiration	Turbocharged & Charge Air Cooled
Displacement	14.9 L
Rated Power & Rated Speed	373 kW & 1800 RPM
Peak Torque	2508 N-m at 1200 RPM
Fuel System	Cummins XPI
EGR System	Cooled High Pressure



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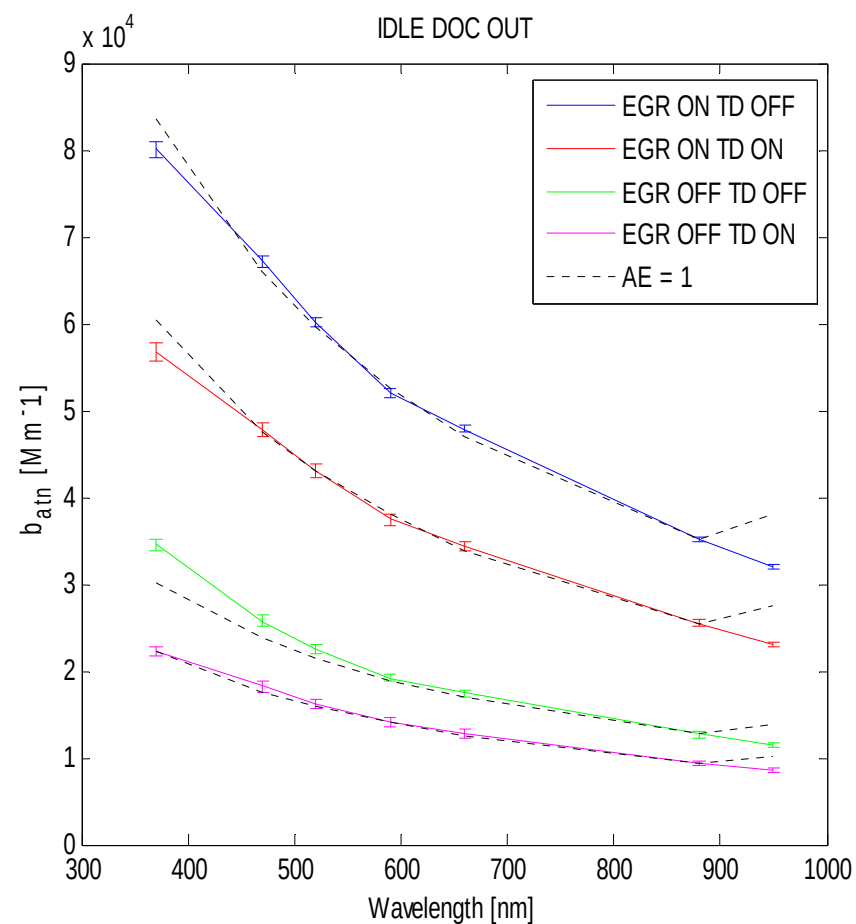
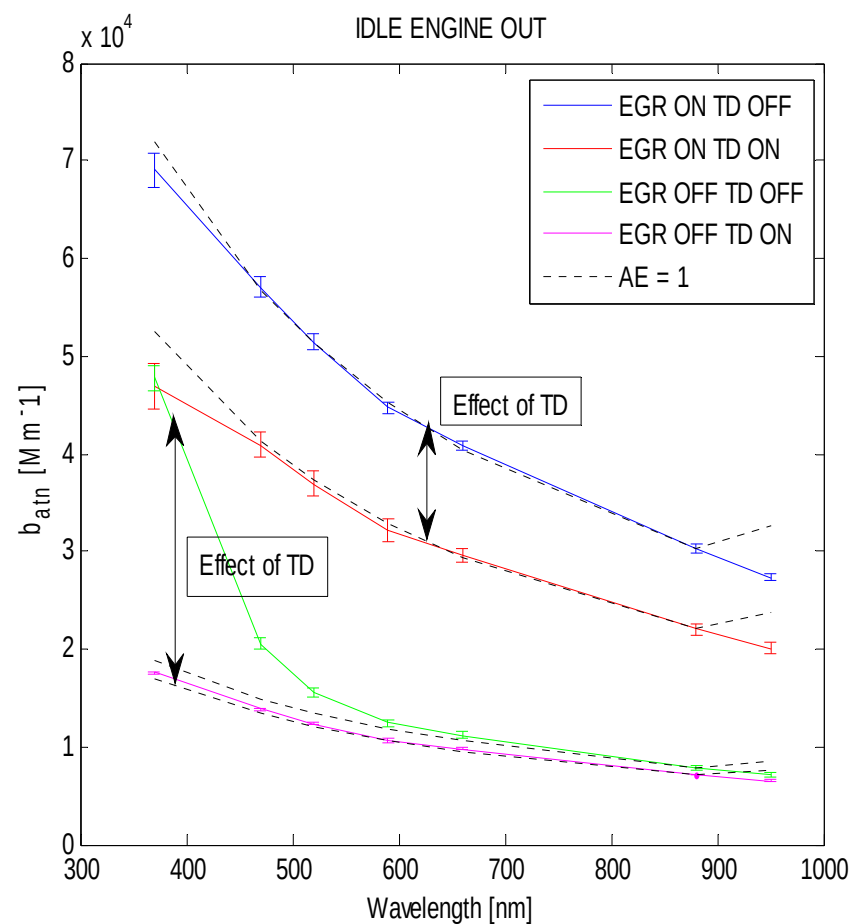
Emissions Testing Lab



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BrC Plots



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Atmospheric Sampling: Objectives

- Conduct field measurements at a variety of sites dominated by various sources of Black Carbon (BC) and Brown Carbon (BrC)
- Determine relative fraction of light absorption by BC and BrC
- Determine sources of BC and BrC
- Develop simple parameterizations for influence of aging on the light absorbing properties of aerosols



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Approach: Specifics

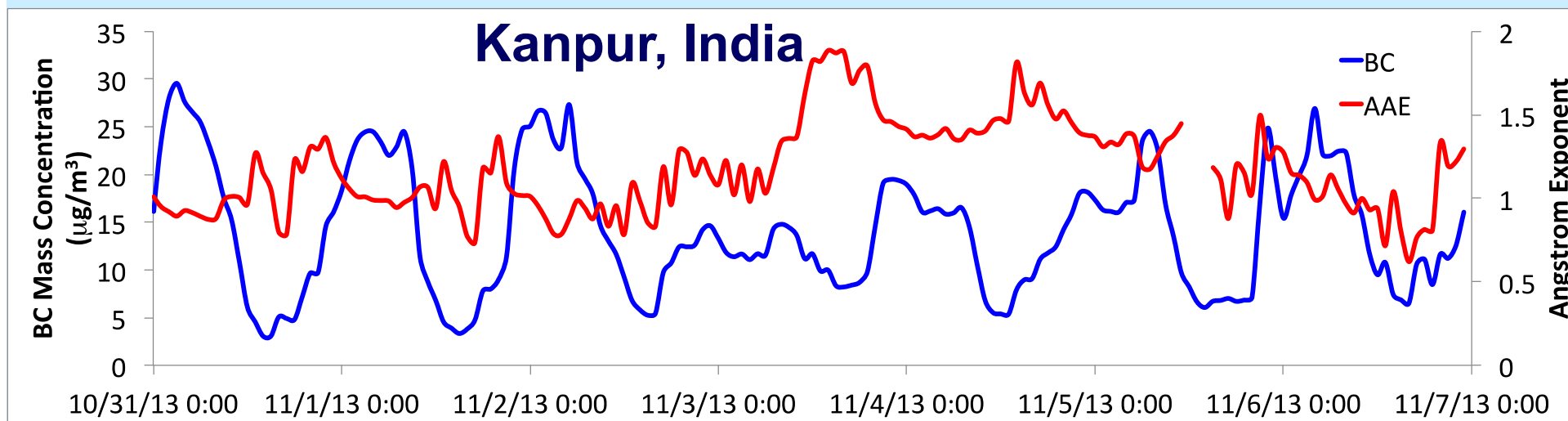
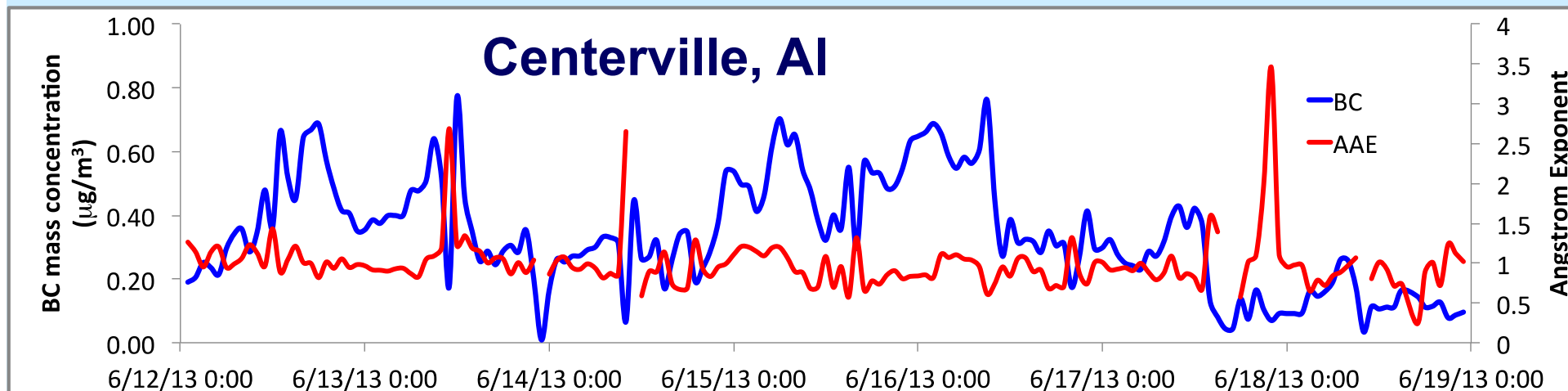
Parameter	Instrument	Dates	Objective
<i>Real-Time Continuous</i>			
$\sigma_{ap}(\lambda)$	Magee Aethalometer, PAX	4-1 Month Periods	Compare with Mie Theory Light Absorption Estimates from MOUDI 1
$\sigma_{sp}(\lambda)$	Radiance Research Nephelometer	--	Compare with Mie Theory Estimates from MOUDI
<i>Time-Integrated Sampling</i>			
EC/OC, Trace Organics, WSOC, $Abs(\lambda)_{solvent}$, $Abs(\lambda)_{water}$	HiVol Filter sampler	4-1 Month Periods	Source apportionment, RI Estimates for Mie Theory, Solvents Extracts for Aerosolization Experiments
EC/OC, WSOC, $Abs(\lambda)_{solvent}$, $Abs(\lambda)_{water}$	MOUDI 1	4-1 Month Periods	Estimation of $\sigma_{ap}(\lambda)$ as function of size for both water and solvent extracts and BC
Mass, Ions	MOUDI 2	4-1 Month Periods	Estimation of $\sigma_{sp}(\lambda)$ as a function of size



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Black Carbon (BC) and Angstrom Absorption Exponent (AAE) in rural US and India



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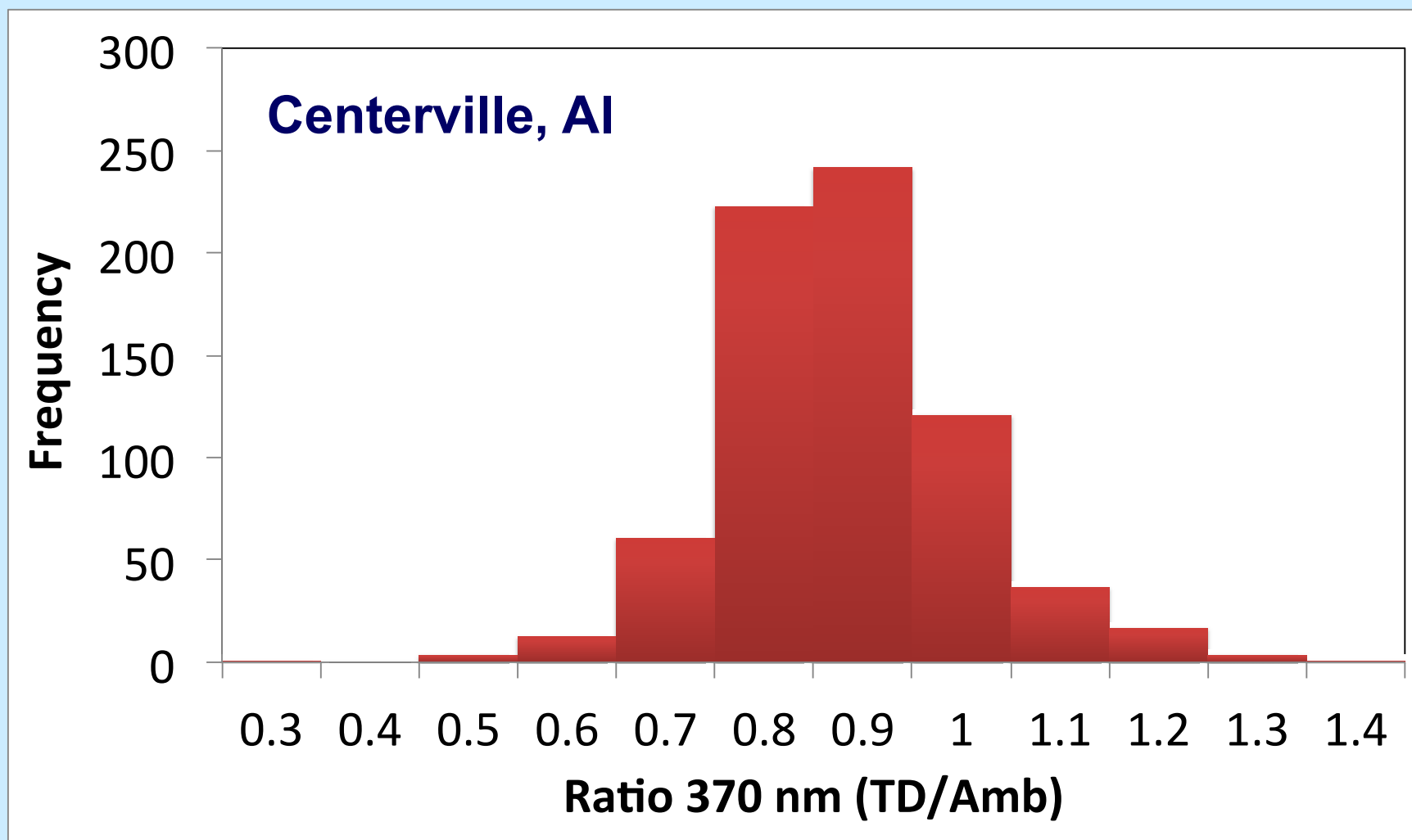
Trash/Refuse Burning: A Source of Brown Carbon



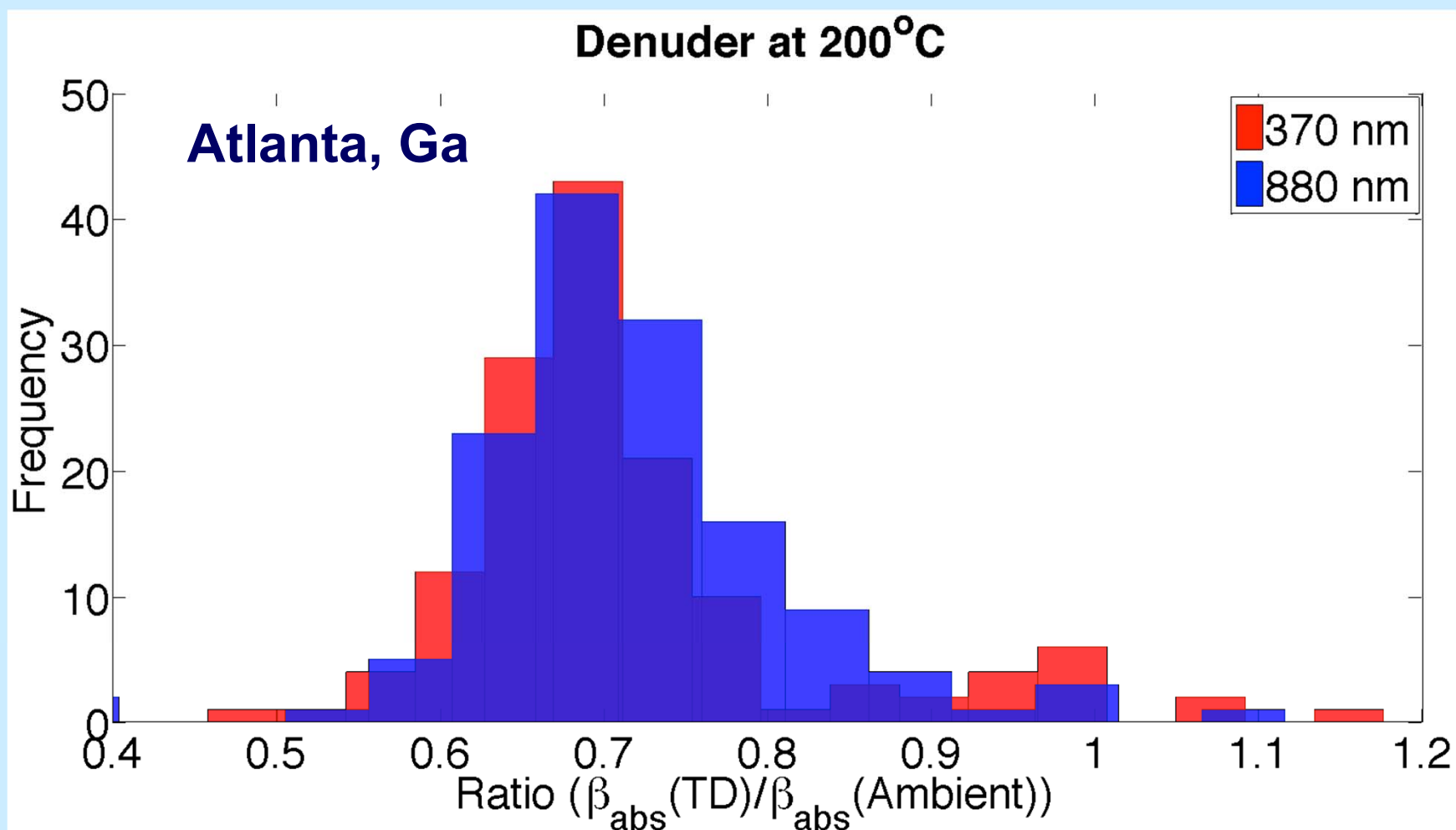
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Ratio of light absorption for denuder (200°C) to ambient air



Ratio of light absorption for denuder (200°C) to ambient air



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Low Cost Sensor Networks



microAet
h-Black
Carbon

Arduino-
microcontroller



PM
sensor

CO₂
Sensor

T, RH



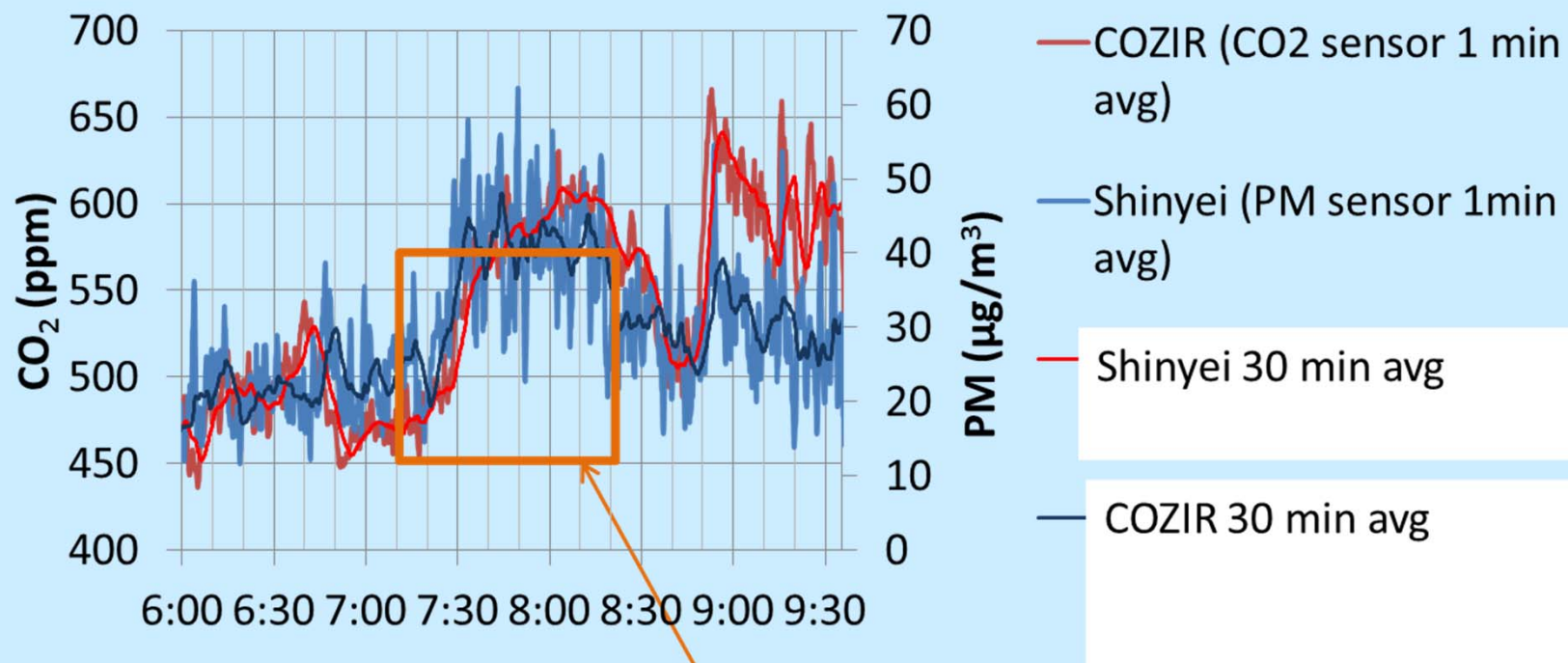
Framework for Black

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Impacts



Atlanta Freeway PM Emission Factor Estimate



Rush Hour Event

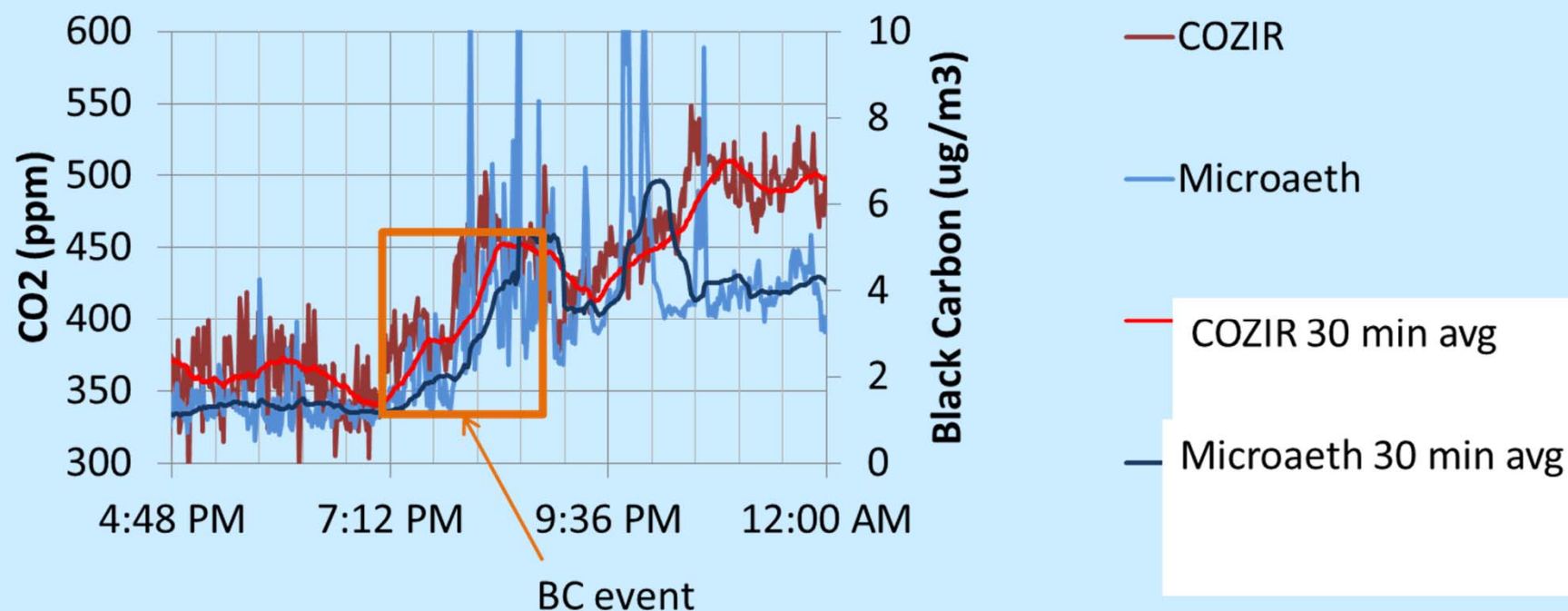
$$\begin{aligned}
 \text{Rough Emissions Factor} &= \Delta \text{PM} / \Delta \text{CO}_2 \\
 &= 0.079 \mu\text{g m}^{-3} \text{ PM/ppmCO}_2 \\
 &= 0.39 \text{ g PM/kg fuel}
 \end{aligned}$$



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Atlanta Freeway BC Emission Factor Estimate



$$\begin{aligned}
 \text{Rough Emissions Factor} &= \Delta \text{BC} / \Delta \text{CO}_2 \\
 &= 0.044 \mu\text{g m}^{-3} \text{ BC/ppmCO}_2 \\
 &= 75 \text{ mg BC/kg fuel}
 \end{aligned}$$



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Ongoing Efforts

- Source Testing
 - Applying methodology to other source of concern: real world biomass, residential coal
- Atmospheric Sampling
 - Water and methanol extractions of size-resolved BC and BrC samples
 - Extraction of hivol samples to determine optical properties and sources of light absorption
- Publications

